

Starting System

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)	
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent
 ST3093-A	Fluke 77-IV multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Principles of Operation

NOTE: This vehicle is equipped with Passive Anti-Theft System (PATS) that disables the engine from cranking and starting if an unprogrammed PATS key is used or an invalid PCM ID is received. PATS is controlled by the Body Control Module (BCM) . If there is a PATS concern that causes the engine to disabled, the Instrument Panel Cluster (IPC) displays "STARTING SYSTEM FAULT" in the message center. Refer to Section 419-01B to diagnose a PATS concern.

The starter system controls the cranking of the engine. The starter motor is enabled by the starter relay when the relay is activated by the PCM . The starting system is electronically controlled by the PCM . The PCM requires the following inputs:

- Ignition switch request to start the engine
- Run/start relay voltage
- PATS enable status from the BCM
- Correct transmission PARK or NEUTRAL input from the Transmission Range (TR) sensor

Under normal operation when the ignition is turned to the START position, the BCM generates a key verification message. It sends this message to the PATS transceiver. The transceiver reads the key and sends a key verification message back to the BCM . If the message received does not match the key verified in the BCM memory, the engine is disabled and does not crank and start. Once the BCM determines the correct programmed key is in the ignition, the BCM sends the PCM a valid PATS message. When the PCM receives the message, it generates a response and sends it back to the BCM . If the response from the PCM does not match the response in the BCM memory, PATS is not enabled and the engine does not crank. If both the key verification and PCM ID pass, PATS is enabled. The PCM recognizes the correct electronically coded passive key and provides voltage and ground to the starter relay coil. The starter relay coil is energized causing the relay contacts to close providing voltage to the starter solenoid, allowing the starter motor to crank and start the engine. The PCM disengages the starter once an engine rpm threshold is reached, a set crank time is exceeded or the ignition is turned OFF. The BCM also enables the Run/start relay which sends voltage to the PCM . If the PCM does not receive this voltage the engine will not crank.

The  sensor prevents operation of the starter motor unless the transmission is in NEUTRAL or PARK.

Remote Start

The remote start system (if equipped) enables the engine to be started from up to 100 meters (328 feet) away, by pressing a sequence of buttons on the Integrated Keyhead Transmitter (IKT) . The factory remote start system does not operate if the ignition is in the RUN/START position or the hood is not closed. An additional external antenna is connected to the Tire Pressure Monitor Module (TPM) to extend the  range. The remote start system can be enabled or disabled by the customer. When enabled, the customer can set the engine run time for 5, 10 or 15 minutes. For information on setting the engine run time, refer to the Owner's Literature or Section 413-01.

The remote start system is activated by pressing the lock button on the , to make sure the doors are locked, followed by 2 presses of the remote start button. The 3 button presses must occur within 3 seconds. The system acknowledges the start request by flashing the exterior lights twice. Once started, the engine runs for 10 minutes (default run time is 10 minutes; the time can be programmed by the customer to 5, 10 or 15 minutes) and the parking lights remain illuminated until the cycle is complete. The remote start duration can also be extended by initiating remote start while the first timer is still running or by initiating a completely independent start cycle when the first has completed. During the remote start cycle the power windows are inhibited and the radio will not turn on automatically but may be switched on manually from inside the vehicle. If the remote start system fails to start the horn chirps twice.

To transition from remote start mode to drive mode, use a programmed  to switch the ignition to the RUN position. If a programmed  is not used, the engine shuts off.

To deactivate the remote start system, press the remote start button once. When deactivating the remote start system the parking lights turn off to give a visual indication that the engine and climate control features have turned off. It may be necessary to be closer to the vehicle to deactivate the remote start feature than it was to initiate it due to ground deflections and added noise from the engine running.

The remote start system conditions the cabin temperature during remote start based on the outside temperature. For information on cabin temperature conditioning during remote start, refer to Section 412-00.

Inspection and Verification

WARNING: Always disconnect the battery ground cable at the battery before disconnecting the starter motor battery terminal lead. If a tool is shorted at the starter motor battery terminal, the tool can quickly heat enough to cause a skin burn. Failure to follow this instruction may result in serious personal injury.

NOTE: Make sure the anti-theft system is functioning correctly before carrying out a logical starting system diagnosis. Address anti-theft system concerns before continuing. Refer to Section 419-01B.

1. Verify the customer concern by operating the starting system.
2. Check the battery for loose, damaged or corroded connections.
3. Check the battery condition and state of charge. Section 414-01.
4. Perform Battery Drain Test and check for abnormal ignition-off current drain(s). Refer to Section 414-01 General Procedures.
5. Remove the accessory drive belt. Refer to Section 303-05. Verify the crankshaft and each of the components driven by the accessory drive belt rotate and are not seized or damaged.
6. If any aftermarket accessories have been added to the vehicle, make sure they are properly wired.
7. Inspect to determine if any of the following mechanical or electrical concerns apply.

Visual Inspection Chart

Mechanical	Electrical
• Starter motor • Starter motor bolts	• Battery • Battery cables • Battery Junction Box (BJB) Starter relay • Battery Junction Box (BJB) Run/start relay • Battery Junction Box (BJB) fuse 13 (30A) • Body Control Module (BCM) fuse 28 (15A) • Loose or corroded connections

8. If an obvious cause for an observed or reported concern is found, correct the cause before proceeding to the next step.

9. **NOTE:** *Make sure to use the latest scan tool software release.*

If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

10. **NOTE:** *The LED prove out confirms power and ground from the DLC are provided to the PCM .*

- If the scan tool does not communicate with the PCM :
- check the PCM connection to the vehicle.
 - check the scan tool connection to the PCM .
 - refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

11. If the scan tool does not communicate with the vehicle:
- verify the ignition key is in the ON position.
 - verify the scan tool operation with a known good vehicle.
 - refer to Section 418-00 to diagnose no communication with the scan tool.

12. Carry out the network test.
- If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
 - If the network test passes, proceed with module self-test diagnostics.

13. Carry out the self-test diagnostics for the BCM , PCM and Tire Pressure Monitor Module (TPM) (TPM is standard equipment, also used for optional remote start feature).

14. If the DTCs retrieved are related to the concern, go to DTC Charts.

15. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

DTC Charts

PCM DTC Chart

DTC	Description	Action
P0512	Starter Request Circuit — Circuit has Power With the Ignition in the OFF Position	CARRY OUT the Ignition Switch Component Test. Refer to Wiring Diagrams Cell 149 for component testing. If necessary, INSTALL a new ignition switch. If the ignition switch passed the component test, REPAIR circuit CDC35 (BU/WH) for a short to voltage.
P0705	Transmission Range Sensor A Circuit (PRNDL Input): No Sub Type Information	REFER to Section 307-01 to diagnose the Transmission Range (TR) sensor concern.
P0706	Transmission Range Sensor "A" Circuit Range/Performance: No Sub	REFER to Section 307-01 to diagnose the <u>TR</u> sensor concern.

DTC	Description	Action
	Type Information	
P0707	Transmission Range Sensor "A" Circuit Low: No Sub Type Information	REFER to Section 307-01 to diagnose the T R sensor concern.
P0708	Transmission Range Sensor "A" Circuit High: No Sub Type Information	REFER to Section 307-01 to diagnose the T R sensor concern.
P0709	Transmission Range Sensor "A" Circuit Intermittent: No Sub Type Information	REFER to Section 307-01 to diagnose the T R sensor concern.
P161A	Incorrect Response From Immobilizer Control Module	REFER to Section 419-01B.
P1702	Transmission Range Sensor Circuit Intermittent: No Sub Type Information	REFER to Section 307-01 to diagnose the T R sensor concern.
P1705	Transmission Range Circuit Not Indicating Park/Neutral During Self Test: No Sub Type Information	REFER to Section 307-01 to diagnose the T R sensor concern.
P1921	Transmission Range Signal: No Sub Type Information	REFER to Section 307-01 to diagnose the T R sensor concern.
P2532	Ignition Switch Run Position Circuit High	GO to Pinpoint Test G.
P2535	Ignition Switch Run/Start Position Circuit High	GO to Pinpoint Test G.
All other PCM DTCs	—	REFER to Section 303-14.

Body Control Module (BCM) DTC Chart

DTC	Description	Action
B1305:01	Hood Switch: General Electrical Failure	REFER to Section 419-01A.
C113A:11	Wake up Control: Circuit Short to Ground	GO to Pinpoint Test C.
C113A:15	Wake up Control: Circuit Short to Battery or Open	GO to Pinpoint Test C.
U1000:00	Solid State Driver Protection Activated - Driver Disabled: No Sub Type Information	The Body Control Module (BCM) has temporarily disabled an output because an excessive current draw exists (such as a short to ground). The BCM cannot enable the output until the cause of the short is corrected. ADDRESS all other DTCs first. After the cause of the concern is corrected, CLEAR the DTCs. REPEAT the self-test.
All other DTCs	—	REFER to Section 419-10.

Tire Pressure Monitor (TPM) Module DTC Chart

DTC	Description	Action
B1D55:01	Antenna #2: General Electrical Failure	GO to Pinpoint Test D.
All other DTCs	—	REFER to Section 204-04.

Symptom Chart

Symptom Chart		
Condition	Possible Causes	Action
The engine does not crank	- Battery - Battery cables - Fuses - Starter motor - PCM - Starter relay - Run/start relay - Ignition switch - Transmission Range (TR) sensor - Passive Anti-Theft System (PATS) - Body Control Module (BCM) - Wiring, terminals or connectors	GO to Pinpoint Test A.
Unusual starter noise	- Starter motor - Starter motor mounting - Starter motor engagement - Flexplate ring gear	GO to Pinpoint Test B.

• The engine cranks slowly	• Battery • Battery cables • Starter motor • Wiring, terminals or connectors	• CARRY OUT the Starter Motor — Feed Circuit and Starter Motor — Ground Circuit Tests. REFER to Component Tests in this section.
• The engine cranks but does not start.	• Fuses • PCM • Fuel pump • Fuel pump relay	• Powertrain Control/Emissions Diagnosis (PC/ED) manual
• The starter spins but the engine does not crank	• Damaged flexplate • Starter motor	• REFER to Starter Motor Drive Gear and Flywheel Ring Gear Inspection in this section to continue diagnosis.
• The starter does not disengage from the flexplate	• Starter motor • Wiring, terminal or connectors • Starter relay	• REFER to Starter Motor Drive Gear and Flywheel Ring Gear Inspection in this section to continue diagnosis. • REMOVE the starter relay, if the engine stops cranking INSTALL a new relay. If the engine continues to crank, REPAIR circuit CDC25 (BN/GN) for a short to voltage.
• The remote start system operates with reduced range	• Wiring, terminals or connectors • External antenna • Tire Pressure Monitor Module (TPM)	• GO to Pinpoint Test D.
• The remote start is inoperative	• Wiring, terminals or connectors • Remote start feature not enabled • Integrated Keyhead Transmitter (IKT) • Hood switch • PCM	• GO to Pinpoint Test E.
• One-Touch Integrated Start (OTIS) is inoperative	• OTIS not enabled	• Using a scan tool, ENABLE OTIS .

Pinpoint Tests

Pinpoint Test A: The Engine Does Not Crank

Refer to Wiring Diagrams Cell [20](#) , Starting System for schematic and connector information.

Normal Operation

When the ignition switch is turned to the START position, the Body Control Module (BCM) and PCM receive a request to start the engine. When the PCM receives the correct coded key ID messages from the BCM , voltage from the Run/start relay, and the correct input from the Transmission Range (TR) sensor that the gear selector is in PARK or NEUTRAL, voltage and ground is supplied to the starter relay coil. When the starter relay is energized the relay contacts close. Voltage is then supplied from Battery Junction Box (BJB) fuse 13 (30A) to the starter relay contacts which supplies voltage to the starter solenoid. The starter solenoid is grounded at the starter motor. Energizing the starter solenoid engages the starter drive into the ring gear and closes the starter solenoid contacts allowing voltage directly from the battery to the starter motor to crank the engine. The PCM disengages the starter once an RPM threshold is reached, a set crank time is exceeded or the ignition switch is turned to the OFF position.

This pinpoint test is intended to diagnose the following:

- Battery
- Battery cables
- Fuses
- Ignition switch
- PCM
- Starter motor
- Starter relay
- TR sensor
- Passive Anti-Theft System (PATS)
- BCM
- Wiring, terminals or connectors

PINPOINT TEST A : THE ENGINE DOES NOT CRANK

A1 PERFORM INSPECTION AND VERIFICATION

NOTE: Make sure battery voltage is greater than 12.2 volts prior to and during this pinpoint test.

NOTE: If it is necessary to leave the battery charger connected during testing, do not leave it on its highest setting. The highest and boost settings can occasionally exceed 16 volts, resulting in false test results and setting DTCs.

- Perform the Inspection and Verification procedure in this section.

Was an obvious cause for an observed or reported concern found?

Yes	Correct the cause as necessary.
No	GO to A2 .

A2 CHECK FOR ANY MESSAGES IN THE MESSAGE CENTER

- Ignition ON.
- Check the message center with the ignition key in the RUN position.

Is STARTING SYSTEM FAULT displayed in the message center?

Yes	REFER to Section 419-01B.
No	GO to A3 .

A3 VERIFY BCM AND PCM PASS THE NETWORK TEST

- Ignition ON.
- Using a diagnostic scan tool, perform the Network Test.

Did the BCM and PCM pass the Network Test?

Yes	GO to A4 .
No	REFER to Section 418-00.

A4 RETRIEVE BCM AND PCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: **BCM** — Self-Test .
- Enter the following diagnostic mode on the scan tool: **PCM** — Self-Test .

Were DTCs retrieved during the self-test?

Yes	REFER to the DTC Charts in this section.
No	GO to A5 .

A5 CHECK THE PCM TRANSMISSION RANGE TR SENSOR PID

- Enter the following diagnostic mode on the scan tool: DataLogger — **PCM** .
- While observing the **TR PID** , place the gear selector in PARK and then NEUTRAL.

Does the PID match the selector lever position?

Yes	GO to A6 .
No	REFER to Section 307-01, to diagnose the TR sensor concern.

A6 CHECK THE BCM IGN_SW_STATE PID

- Enter the following diagnostic mode on the scan tool: DataLogger — **BCM** .
- While holding the key in the START position, monitor the **BCM IGN_SW_STATE PID** .

Does the PID change from OFF to START?

Yes	GO to A7 .
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No	DIAGNOSE No power in Start. REFER to Section 211-05.
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A7 CHECK THE STARTER MOTOR RELAY

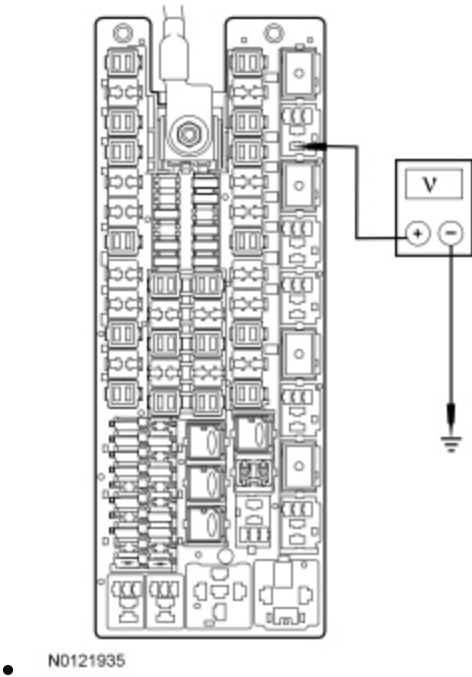
- Swap the starter relay with a known good relay and attempt to crank the engine.

Does the engine crank?

Yes	INSTALL a new starter relay.
No	GO to A8 .

A8 CHECK THE VOLTAGE TO THE STARTER RELAY

- Ignition OFF.
- Disconnect: **BJB** Starter Relay .
- Measure the voltage between starter relay socket 3, circuit SBB13 (GY/RD) and ground.

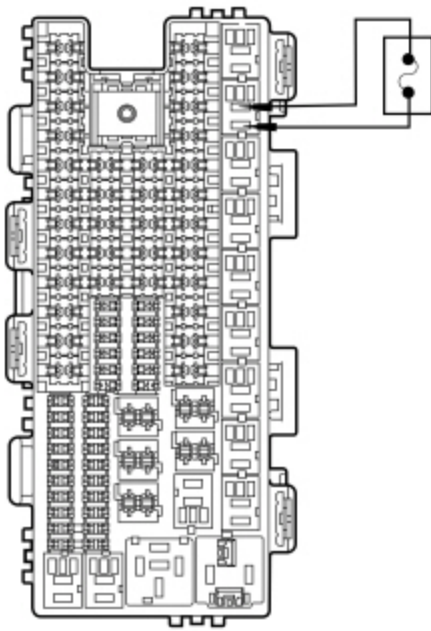


Is the voltage greater than 11 volts?

Yes	GO to A9 .
No	VERIFY BJB fuse 13 (30A) is OK. If OK, REPAIR circuit SBB13 (GY/RD) for an open. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. TEST the system for normal operation.

A9 CHECK THE STARTER MOTOR

- With the transmission in PARK, connect one end of a fused jumper wire to starter relay socket 3, circuit SBB13 (GY/RD) and momentarily connect the other end of the fused jumper wire to starter relay socket 5, circuit CDC25 (BN/GN).



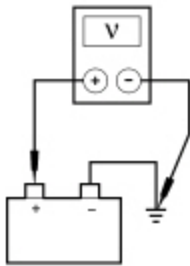
• N0121425

Did the starter motor engage and the engine crank?

Yes	GO to A15 .
No	GO to A10 .

A10 CHECK THE BATTERY GROUND CABLE

- Measure the voltage between the positive battery post and the battery ground cable connection at the engine.



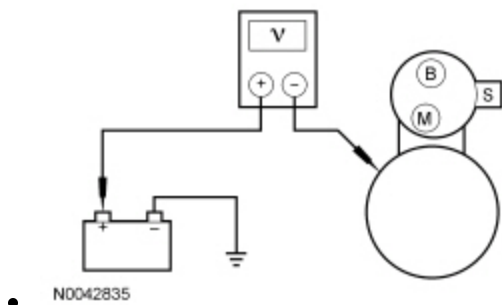
• AJ0280-A

Is the voltage greater than 11 volts?

Yes	GO to A11 .
No	INSTALL new battery cables. REFER to Section 414-01. TEST the system for normal operation.

A11 CHECK THE STARTER MOTOR GROUND

- Measure the voltage between the positive battery post and the starter motor case.

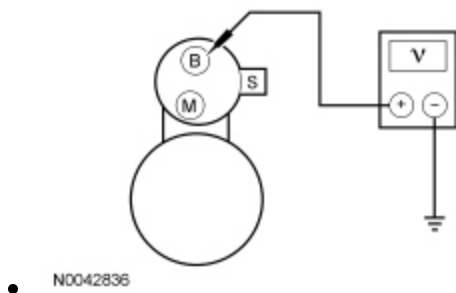


Is the voltage greater than 11 volts?

Yes	GO to A12 .
No	CLEAN the starter motor mounting flange and make sure the starter motor is correctly mounted. TEST the system for normal operation.

A12 CHECK THE VOLTAGE TO THE STARTER MOTOR

- Measure the voltage between the starter solenoid [C197A](#) Pin 1, circuit SDC02 (RD) and ground.

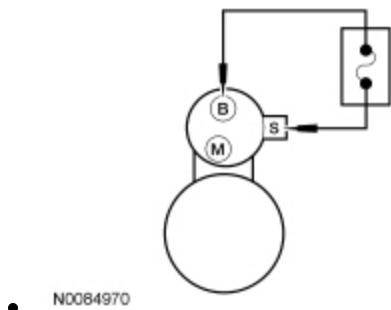


Is the voltage greater than 11 volts?

Yes	GO to A13 .
No	INSTALL new battery cables. REFER to Section 414-01. TEST the system for normal operation.

A13 CHECK THE STARTER MOTOR

- Connect one end of a remote starter switch to starter solenoid [C197A](#) Pin 1, circuit SDC02 (RD) and the other end of the remote starter switch to starter solenoid [C197B](#) Pin 1, circuit CDC25 (BN/GN).



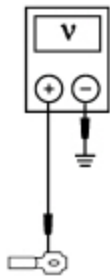
- Engage the remote starter switch momentarily.

Did the starter engage and the engine crank?

Yes	Remove the remote starter switch and GO to A14 .
No	INSTALL a new starter motor. REFER to Starter Motor — 3.5L GTDI, 3.7L or Starter Motor — 5.0L (4V), 6.2L (2V) in this section. TEST the system for normal operation.

A14 CHECK FOR START INPUT AT THE STARTER

- Connect: ~~BJB~~ Starter Relay .
- Disconnect: Starter Solenoid [C197B](#).
- Measure the voltage between the starter solenoid [C197B](#) Pin 1, circuit CDC25 (BN/GN) and ground, while holding the key in the START position.



AJ0285-A

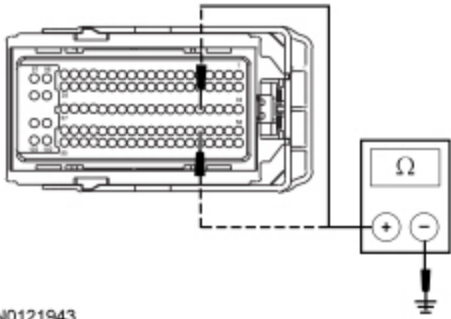
Is the voltage greater than 11 volts?

Yes	CLEAN the starter solenoid "S" terminal and C197B . CHECK the wiring and the starter motor for a loose or intermittent connection. TEST the system for normal operation.
No	REPAIR circuit CDC25 (BN/GN) for an open. TEST the system for normal operation.

A15 CHECK THE PCM START CIRCUITS FOR A SHORT TO GROUND

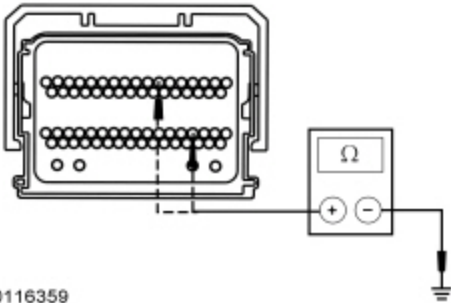
- Disconnect: PCM [C1551B](#) (3.5L) [C175B](#) (3.7L) or [C1381B](#) (5.0L and 6.2L) .
- Measure the resistance between the PCM [C1551B](#) (3.5L), harness side and ground, using the following chart:

PCM Connector-Pin	Circuit
C1551B Pin 43 (3.5L)	CDC12 (YE)
C1551B Pin 62 (3.5L)	CDC54 (WH/GN)



- N0121943
- Measure the resistance between the PCM [C175B](#) (3.7L) or [C1381B](#) (5.0L and 6.2L), harness side and ground, using the following chart:

PCM Connector-Pin	Circuit
C175B Pin 7 (3.7L) or C1381B Pin 7 (5.0L and 6.2L)	CDC12 (YE)
C175B Pin 37 (3.7L) or C1381B Pin 37 (5.0L and 6.2L)	CDC54 (WH/GN)



• N0116359

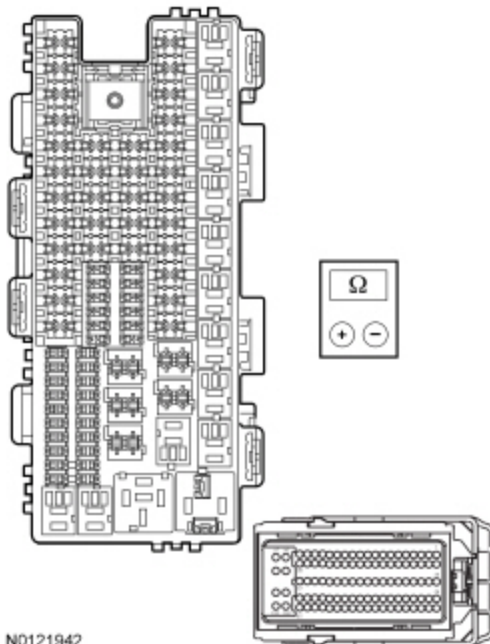
Are the resistances greater than 10,000 ohms?

Yes	GO to A16 .
No	REPAIR the affected circuit. TEST the system for normal operation.

A16 CHECK THE PCM START CIRCUITS FOR AN OPEN

- Measure the resistance between the PCM [C1551B](#) (3.5L), harness side and the starter relay using the following chart:

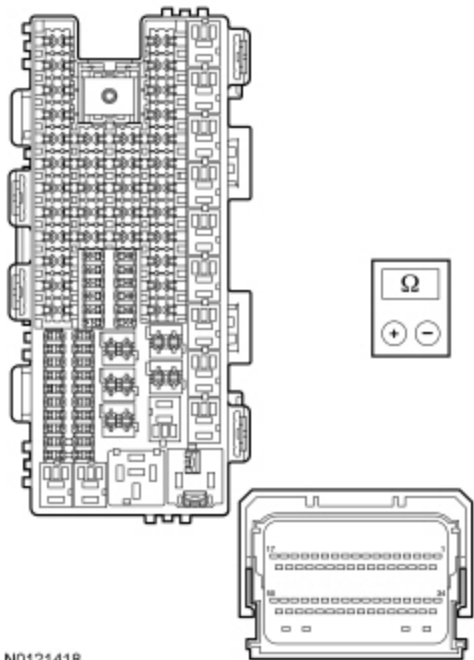
PCM Connector-Pin	Circuit	Starter Relay Socket
C1551B Pin 43 (3.5L)	CDC12 (YE)	2
C1551B Pin 62 (3.5L)	CDC54 (WH/GN)	1



N0121942

- Measure the resistance between the PCM [C1381B](#) (5.0L and 6.2L) or [C175B](#) (3.7L), harness side and the starter relay using the following chart:

PCM Connector-Pin	Circuit	Starter Relay Socket
C175B Pin 7 (3.7L) or C1381B Pin 7 (5.0L and 6.2L)	CDC12 (YE)	2
C175B Pin 37 (3.7L) or C1381B Pin 37 (5.0L and 6.2L)	CDC54 (WH/GN)	1



N0121418

Are the resistances less than 5 ohms?

Yes	INSTALL a new PCM . REFER to Section 303-14. TEST the system for normal operation.
No	REPAIR the affected circuit. TEST the system for normal operation.

Pinpoint Test B: Unusual Starter Noise

Normal Operation

Correct starter operation relies on correct mounting of the starter to the engine, alignment of the starter ring gear to the flexplate and correct functioning of the starter assembly (internal gears, bearings).

This pinpoint test is intended to diagnose the following:

- Starter motor
- Starter motor mounting
- Starter motor engagement
- Flexplate ring gear

PINPOINT TEST B : UNUSUAL STARTER NOISE

B1 CHECK THE STARTER MOUNTING

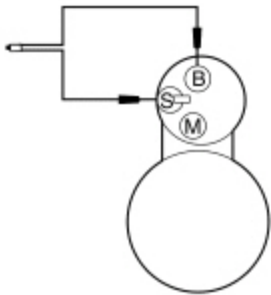
- Inspect the starter motor mounting bolts for looseness.

Is the starter motor mounted correctly?

Yes	GO to B2 .
No	INSTALL the starter motor correctly. REFER to Starter Motor — 3.5L GTDI, 3.7L or Starter Motor — 5.0L (4V), 6.2L (2V) in this section. TEST the system for normal operation.

B2 CHECK FOR STARTER NOISE

- Ignition OFF.
- Connect a remote starter switch between the starter solenoid [C197A](#) Pin 1 and [C197B](#) Pin 1.



- Engage the starter and verify the noise is due to starter operation.

Is the noise due to the starter engagement?

Yes	REFER to Starter Motor Drive Gear and Flywheel Ring Gear Inspection in this section to continue diagnosis.
No	REFER to Section 303-00, Symptom Chart — Engine NOISE to continue diagnosis.

Pinpoint Test C: DTC C113A:11 or C113A:15

Refer to Wiring Diagrams Cell [112](#) , Passive Anti-Theft System for schematic and connector information.

Normal Operation

The wake up control circuit is used to wake up the PCM prior to engine cranking. The PCM needs to wake up prior to a crank request so that it has time to go through its own initialization. The wake up control circuit also powers the Passive Anti-Theft System (PATS) transceiver. The wake up control circuit is controlled by the Body Control Module (BCM) and receives voltage from fuse 18 (10A). The BCM uses a Field-Effect Transistor (FET) to control the wake up control circuit. A DTC sets and the wake up control circuit is disabled via the FET if a fault is detected. Once the condition that caused the DTC to set is repaired, the BCM must pass a self-test in order for the DTC to clear and the system to resume normal operation.

The BCM activates the wake up control circuit when:

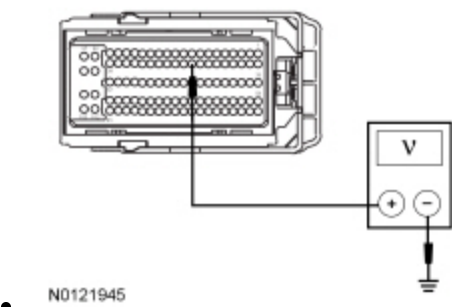
- the driver door is opened.
- a remote start request is received (if equipped with factory remote start).
- the brake pedal is pressed.
- a key is inserted into the ignition.
- the ignition is in the RUN or START position.
- DTC C113A:11 (Wake up Control: Circuit Short To Ground) — sets when the BCM detects a short to ground on the wake up control circuit.
- DTC C113A:15 (Wake up Control: Circuit Short To Battery or Open) — sets when the BCM detects a short to voltage or an open on the wake up control circuit.

This pinpoint test is intended to diagnose the following:

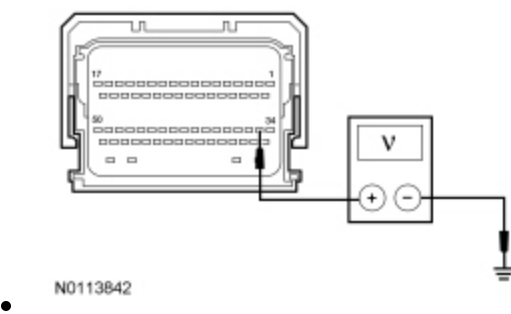
- Wiring, terminals or connectors
- PATS transceiver
- PCM
- BCM

PINPOINT TEST C : DTC C113A:11 OR C113A:15	
NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.	
C1 RETRIEVE THE BCM DTCS	
• Ignition ON. • Enter the following diagnostic mode on the scan tool: BCM — Self-Test .	
Is DTC C113A:15 on demand during the self-test?	
Yes	GO to C2 .
No	GO to C5 .
C2 CHECK THE WAKE UP CONTROL CIRCUIT FOR A SHORT TO VOLTAGE	
• Ignition OFF. • Disconnect: BCM C2280F .	

- Disconnect: PATS Transceiver [C252](#).
- Disconnect: PCM [C1551B](#) (3.5L) [C175B](#) (3.7L) or [C1381B](#) (5.0L and 6.2L) .
- Ignition ON.
- Measure the voltage between the PCM [C1551B](#) Pin 28 (3.5L), circuit CE436 (VT/OG), harness side and ground.



- Measure the voltage between the PCM [C175B](#) Pin 35 (3.7L) or [C1381B](#) Pin 35 (5.0L and 6.2L), circuit CE436 (VT/OG), harness side and ground.

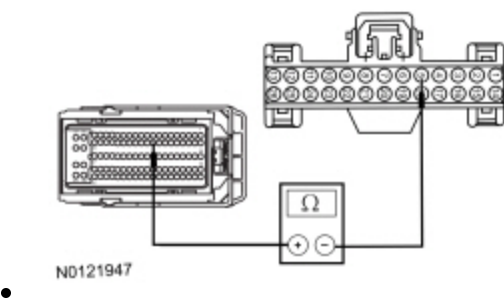


Is any voltage present?

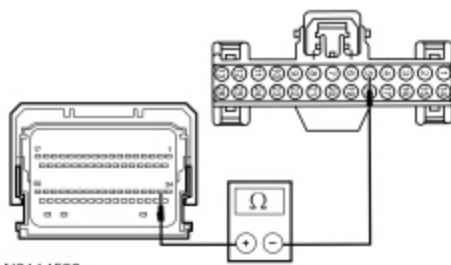
Yes	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	GO to C3 .

C3 CHECK THE WAKE UP CONTROL CIRCUIT FOR AN OPEN

- Ignition OFF.
- Measure the resistance between the PCM [C1551B](#) Pin 28 (3.5L), circuit CE436 (VT/OG), harness side and BCM [C2280F](#) Pin 5, circuit CE436 (VT/OG), harness side.



- Measure the resistance between the PCM [C175B](#) Pin 35 (3.7L) or [C1381B](#) Pin 35 (5.0L and 6.2L), circuit CE436 (VT/OG), harness side and BCM [C2280F](#) Pin 5, circuit CE436 (VT/OG), harness side.



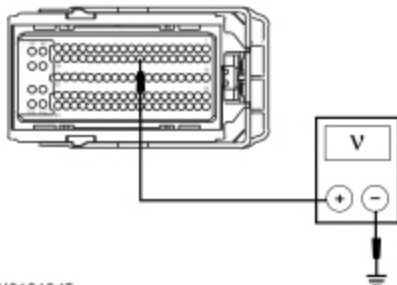
N0114560

Is the resistance less than 5 ohms?

Yes	GO to C4 .
No	REPAIR circuit CE436 (VT/OG) for an open. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

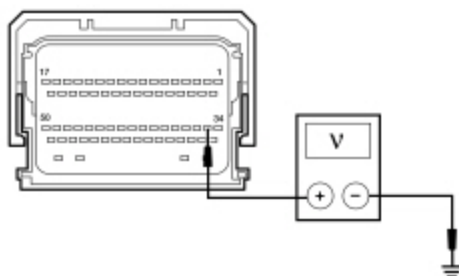
C4 CHECK THE WAKE UP SUPPLY FOR CORRECT OPERATION

- Connect: BCM [C2280F](#) .
- Ignition ON.
- Measure the voltage between PCM [C1551B](#) Pin 28 (3.5L), circuit CE436 (VT/OG), harness side and ground.



N0121945

- Measure the voltage between the PCM [C175B](#) Pin 35 (3.7L) or [C1381B](#) Pin 35 (5.0L and 6.2L), circuit CE436 (VT/OG), harness side and ground.



N0113842

Is the voltage greater than 11 volts?

Yes	INSTALL a new PCM . REFER to Section 303-14. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
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No	VERIFY BCM fuse 18 (10A) is OK. If OK, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
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C5 CHECK THE WAKE UP CONTROL CIRCUIT DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED)

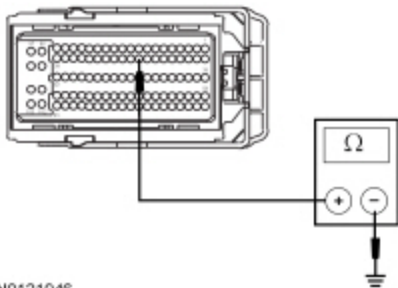
- Enter the following diagnostic mode on the scan tool: Clear BCM Continuous DTCs .
- Ignition OFF.
- Disconnect: PATS Transceiver [C252](#).
- Connect: PCM [C1551B](#) (3.5L) [C175B](#) (3.7L) or [C1381B](#) (5.0L and 6.2L) .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: BCM — Self-Test .

Was DTC C113A:11 retrieved on demand during self-test?

Yes	GO to C6 .
No	GO to C7 .

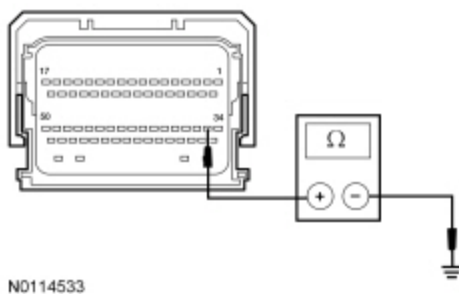
C6 CHECK THE WAKE UP CONTROL CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: BCM [C2280F](#) .
- Measure the resistance between the PCM [C1551B](#) Pin 28 (3.5L), circuit CE436 (VT/OG), harness side and ground.



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- Measure the resistance between the PCM [C175B](#) Pin 35 (3.7L) or [C1381B](#) Pin 35 (5.0L and 6.2L), circuit CE436 (VT/OG), harness side and ground.



N0114533

Is the resistance greater than 10,000 ohms?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

C7 CHECK THE PCM FOR CORRECT OPERATION

- Enter the following diagnostic mode on the scan tool: Clear BCM Continuous DTCs .
- Ignition OFF.
- Connect: PCM C1551B (3.5L) C175B (3.7L) or C1381B (5.0L and 6.2L) .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: BCM — Self-Test .

Was DTC C113A:11 retrieved on-demand during self-test?

Yes	INSTALL a new PCM . REFER to Section 303-14. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	INSTALL a new PATS transceiver. REFER to Section 419-01B. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test D: Poor Remote Start Range

Normal Operation

Vehicles with remote start have an additional external antenna connected to the Tire Pressure Monitor Module (TPM) to extend the RKE range. The Tire Pressure Monitor Module (TPM) receives a signal from the Integrated Keyhead Transmitter (IKT) to enable activation of the remote start function from a distance of up to 100 meters (328 feet).

- DTC B1D55:01 (Antenna #2: General Electrical Failure) — this DTC is set if the vehicle is equipped with remote start but the system feedback circuit does not detect an external antenna.

This pinpoint test is intended to diagnose the following:

- Integrated Keyhead Transmitter (IKT)
- External antenna

- External antenna lead
- TPM

PINPOINT TEST D : POOR REMOTE START RANGE

NOTE: Aftermarket or dealer-installed systems can adversely affect the RKE system operation. These systems must be disconnected before diagnosing any RKE concerns.

D1 CHECK FOR THE CORRECT RKE TRANSMITTERS

- **NOTE:** Make sure the RKE transmitters are those provided with the OEM system and not from an aftermarket system, or a dealer-installed system that may have been installed on the vehicle.

Check that the correct IKTs are used with the vehicle. Compare the IKTs with the one shown in Section 501-14 in the Description and Operation.

Are the correct RKE transmitters present?

Yes	GO to D2 .
No	The system cannot be tested without the correct <u>RKE</u> transmitters. INFORM the customer that the correct <u>RKE</u> transmitters must be present to proceed with diagnosis of the system.

D2 CHECK ALL RKE TRANSMITTERS FOR POOR RANGE PERFORMANCE

- **NOTE:** The 100 m (328 ft) measurement of range is not the standard but is a guideline that clearly indicates a vehicle is experiencing normal range performance.

Check all RKE transmitters for poor range performance (less than 100 m [328 ft]).

Do all RKE transmitters experience poor range?

Yes	GO to D4 .
No	GO to D3 .

D3 CHECK THE IKT BATTERY

- Using a thin coin, open the weak IKT .
- Do not clean off any grease from the battery terminals on the back surface of the circuit board.
- Verify the correct battery is used (CR2032).
- Remove the IKT battery and measure the voltage.

Is the voltage greater than 2.5 volts?

Yes	REPLACE the inoperative <u>IKT</u> and PROGRAM the new <u>IKT</u> individually or PROGRAM all the <u>IKT</u> s . REFER to Section 419-01B for the programming procedures. If all the <u>IKT</u> s are to be programmed, INFORM the customer that any <u>IKT</u> s not present need to be programmed. TEST the system for normal operation.
No	INSTALL a new battery (make sure the battery is seated correctly). DO NOT reprogram the <u>IKT</u> (weak or dead batteries do not erase <u>TIC</u> s from memory). TEST the system for normal operation.

D4 CHECK THE LOCATION OF THE VEHICLE AND THE APPROACH ANGLES AROUND THE VEHICLE

- Make sure the poor performance is consistent in nature and is not from one approaching angle.
- The RKE transmitter range performance may be degraded in certain locations. For example, if the vehicle is within 0.5 mile (0.8 km) of high-power devices or radio/TV towers, the operating distance of the RKE transmitters may be reduced.

Is the poor range performance consistent around the vehicle?

Yes	GO to D5 .
No	The system is operating correctly at this time. The range may have been affected by environmental interference.

D5 CHECK THE EXTERNAL ANTENNA CONNECTION

- Check the connections at the external antenna and TPM .

Are the connections loose or disconnected?

Yes	REPAIR as necessary. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	GO to D6 .

D6 CHECK THE EXTERNAL ANTENNA

- INSTALL a new external antenna.
- ATTEMPT to start the vehicle using the IKT from a distance of 100 meters (328 feet) from the vehicle.

Does the vehicle start from this distance?

Yes	System is operating correctly at this time. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	GO to D7 .

D7 CHECK THE EXTERNAL ANTENNA CABLE

- INSTALL a new external antenna cable. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
- ATTEMPT to start the vehicle using the IKT from a distance of 100 meters (328 feet) from the vehicle

Does the vehicle start from this distance?

Yes	System is operating correctly at this time. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	INSTALL a new TPM . REFER to Section 204-04. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint test E: The Remote Start Is Inoperative

Normal Operation

The factory equipped remote start system enables the engine to be started from up to 100 meters (328 feet) away by pressing the Integrated Keyhead Transmitter (IKT) lock button followed by 2 presses of the remote start button. The 3 button presses must occur within 3 seconds. The IKT signal is received through an external antenna attached to the Tire Pressure Monitor Module (TPM) module and communicated over the Medium Speed Controller Area Network (MS-CAN) to initiate the vehicle start.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Remote start feature disabled
- IKT
- Hood switch
- PCM

PINPOINT TEST E : THE REMOTE START IS INOPERATIVE

E1 CHECK THE OPERATION OF THE IKT	
• Press all buttons on the IKT and check for the correct operation.	
Is the remote start the only function that is inoperative?	
Yes	GO to E2 .
No	REFER to Section 501-14 to DIAGNOSE and REPAIR the Remote Keyless Entry (RKE) system.
E2 CHECK THE VEHICLE STARTING	
• Attempt to start the vehicle.	
Did the vehicle start?	
Yes	GO to E3 .
No	This is not a remote start issue. To diagnose the no crank condition GO to Pinpoint Test A
E3 CHECK FOR ILLUMINATED ENGINE WARNING LAMPS	
• With the engine running check the instrument cluster for illuminated warning lamps.	
Are any engine warning lamps illuminated?	

Yes	CARRY OUT a self test of the PCM . DIAGNOSE and REPAIR any present DTCs. REFER to Section 303-14 or Powertrain Control/Emissions Diagnosis (PC/ED) manual.
No	GO to E4 .

E4 RETRIEVE BCM AND TPM DTCS

- Enter the following diagnostic mode on the scan tool: **BCM** — Self-Test .
- Enter the following diagnostic mode on the scan tool: **TPM** — Self-Test .

Are any DTCs present?

Yes	For BCM DTCs C113A:11 or C113A:15, GO to Pinpoint Test C . For BCM DTC B1305:01, REFER to Section 419-01A. For TPM DTCs U0140:87 or U0422:68, REFER to Section 204-04. For all other DTCs, REFER to Section 419-10.
No	GO to E5 .

E5 CONFIRM THE REMOTE START FUNCTION IS ENABLED

- Ignition ON.
- Check the message center and confirm the remote start function is enabled. Refer to the Owner's Literature or Section 413-01.

Is the remote start function enabled?

Yes	GO to E6 .
No	ENABLE the remote start function. TEST the system for normal operation.

E6 CHECK THE BCM LAST RECEIVED RKE TRANSMITTER BUTTON PRESS (LST_RKE_XMTR) TRANSMITTER PID

- Enter the following diagnostic mode on the scan tool: DataLogger — **BCM** .
- Press the remote start button on the **IKT** and note the **PID** status.

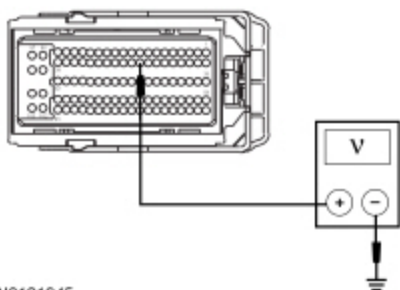
Does the selected **PID** indicate the remote start button was the last button pressed?

Yes	GO to E7 .
No	INSTALL and PROGRAM a new IKT . REFER to Section 419-01B. TEST the system for normal operation.

E7 CHECK THE PCM WAKE UP CIRCUIT

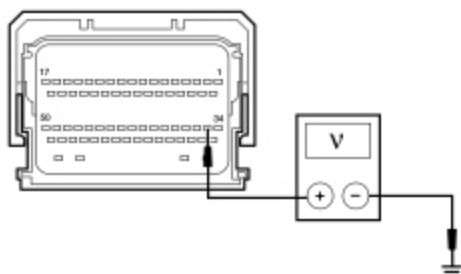
- Ignition OFF.
- Disconnect: **PCM** [C1551B](#) (3.5L) [C175B](#) (3.7L) or [C1381B](#) (5.0L and 6.2L) .

- Ignition ON.
- Measure the voltage between the PCM [C1551B](#) Pin 28 (3.5L), circuit CE436 (VT/OG), harness side and ground.



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- Measure the voltage between the PCM [C175B](#) Pin 35 (3.7L) or [C1381B](#) Pin 35 (5.0L and 6.2L), circuit CE436 (VT/OG), harness side and ground.



N0113842

Is the voltage greater than 11 volts?

Yes	GO to E8 .
No	VERIFY BCM fuse 18 (10A) is OK. If OK, REPAIR circuit CE436 (VT/OG) for an open. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

E8 CHECK THE REMOTE START PRECONDITIONS FOR CORRECT OPERATION

NOTE: The Body Control Module (BCM) acknowledges the remote start request and that all preconditions have been met by providing 2 short flashes of the turn indicators and a system complete chirp.

- Make sure that the following preconditions for remote start have been met:
 - Hood is closed with no DTC present.
 - Ignition must be off
 - Vehicle must be in PARK.
 - Remote start feature is configured ON.
 - Number of remote start events does not exceed allowed time (2 events in a 1 hour duration).
 - Attempt to carry out a remote start.

Is the remote start system still inoperative?

Yes	INSTALL a new PCM . REFER to Section 303-14. TEST the system for normal operation.
No	CARRY OUT the required precondition as necessary. TEST the system for normal operation.

Pinpoint Test G: DTC P2532, P2535

Refer to Wiring Diagrams Cell 20 , Starting System for schematic and connector information.

Normal Operation

When the ignition key is turned to the RUN or START position, the ignition switch supplies voltage to the PCM and BCM . The BCM then grounds the BJB RUN/START relay control circuit to activate the RUN/START relay. When the relay contacts close voltage is supplied to the PCM through Battery Junction Box (BJB) fuse 53 (5A).

- DTC P2532 (Ignition Switch Run Position Circuit High) — sets when the PCM detects a voltage from the ACC/RUN bus when the ignition key is not in the START or RUN position.
- DTC P2535 (Ignition Switch Run/Start Position Circuit High) — sets when the PCM detects a voltage from the RUN/START bus when the ignition key is not in the START or RUN position.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Starter relay
- BJB
- BCM
- PCM

PINPOINT TEST G : DTC P2532, P2535

G1 CHECK THE PCM FOR DTCS

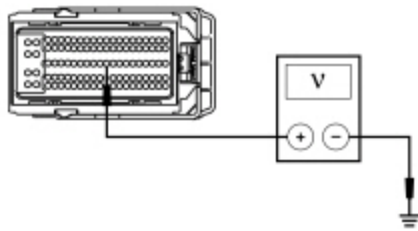
- Ignition ON.
- Check the PCM for DTCS.

Are any DTCS present?

Yes	For DTC P2532,GO to G2. For DTC P2535,GO to G3. For all other PCM DTCS, REFER to Section 303-14.
No	The system is operating correctly at this time. CLEAR the DTCS. REPEAT the self-test. TEST the system for normal operation.

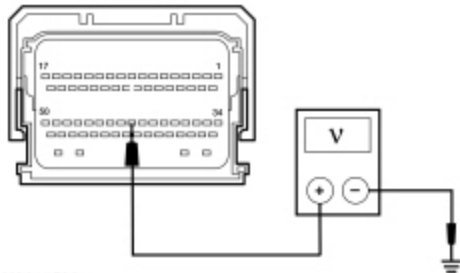
G2 CHECK THE RUN START CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
 - Disconnect: PCM C1551B (3.5L) C175B (3.7L) or C1381B (5.0L and 6.2L) .
 - **NOTE:** Voltage should be present *ONLY* when the ignition switch is in the RUN or START position.
- For 3.5L equipped vehicles, measure the voltage between the PCM C1551B Pin 48, circuit CBB53 (GY/BN), harness side and ground.



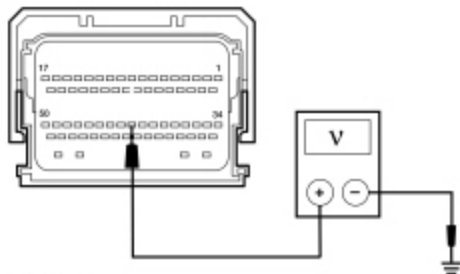
• N0146611

- **For 3.7L equipped vehicles**, measure the voltage between PCM [C175B](#) Pin 42, circuit CBB53 (GY/BN), harness side and ground



• N0114532

- Ignition ON.
- **For 5.0L and 6.2L equipped vehicles**, measure the voltage between PCM [C1381B](#) Pin 42, circuit CBB53 (GY/BN), harness side and ground



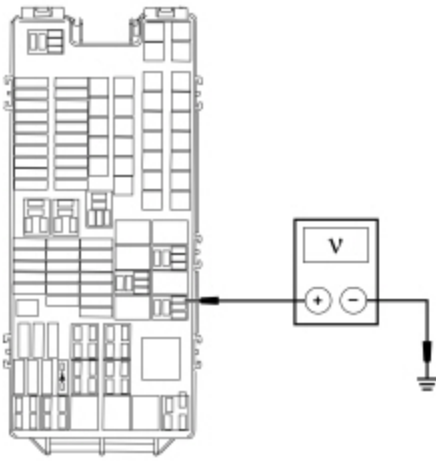
• N0114532

Is any voltage present?

Yes	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. ACTIVATE other systems in the same wire harness. REPAIR any concern found. REPEAT the self-test. TEST the system for normal operation. If the DTC returns. INSTALL a new PCM . REFER to Section 303-14.

G3 CHECK THE START MOTOR REQUEST CIRCUIT

- Start the engine.
- Disconnect: [B1JB](#) Starter Relay .
- Measure the voltage between [B1JB](#) starter relay socket pin 2, circuit CDC26 (GY/OG) and ground.



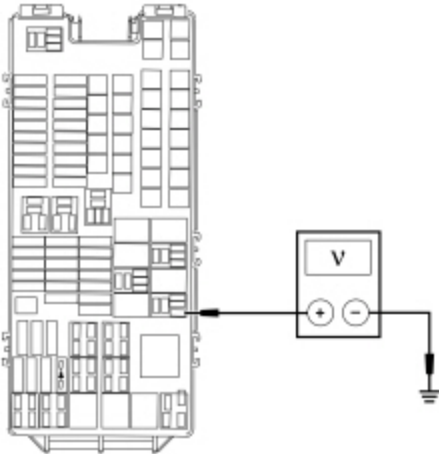
• N0146535

Is any voltage present?

Yes	GO to G4 .
No	INSTALL a new PCM . REFER to Section 303-14. TEST the system for normal operation.

G4 CHECK THE START MOTOR REQUEST CIRCUIT FOR A SHORT VOLTAGE

- Ignition OFF.
- Disconnect: PCM [C1551B](#) (3.5L) [C175B](#) (3.7L) or [C1381B](#) (5.0L and 6.2L) .
- Ignition ON.
- Measure the voltage between [B1B](#) starter relay socket pin 1, circuit CDC12 (YE) and ground.



• N0146536

Is any voltage present?

Yes	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	INSTALL a new PCM . REFER to Section 303-14. TEST the system for normal operation.

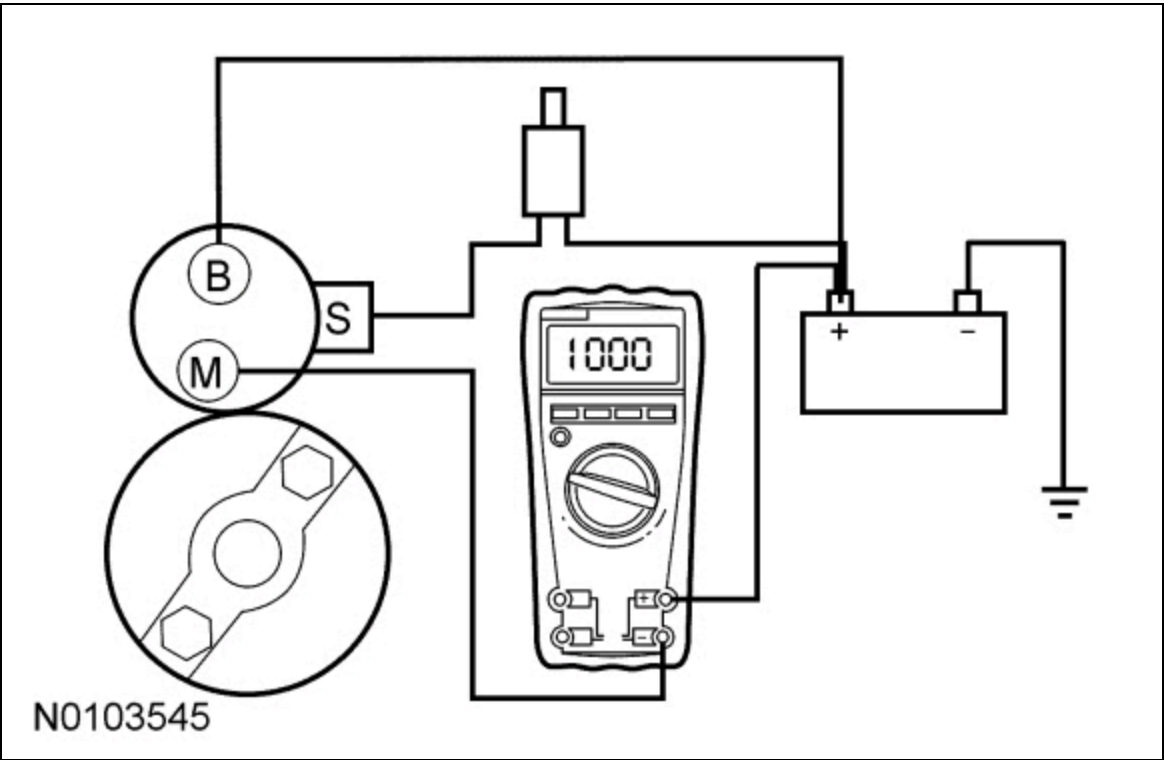
Component Tests

WARNING: Always disconnect the battery ground cable at the battery before disconnecting the starter motor battery terminal lead. If a tool is shorted at the starter motor battery terminal, the tool can quickly heat enough to cause a skin burn. Failure to follow this instruction may result in serious personal injury.

Always make the Fluke 77-IV multimeter connections at the component terminal rather than at the wiring end connector. Making a connection at the wiring end connector could result in false readings because the meter will not pick up a high resistance between the wiring connector and the component.

Starter Motor — Feed Circuit Test

- 1. Make sure the battery is fully charged. Section 414-01.
- 2. Perform a battery drain test. Section 414-01.
- 3. Connect a remote starter switch between starter solenoid “S” terminal and the battery positive terminal.



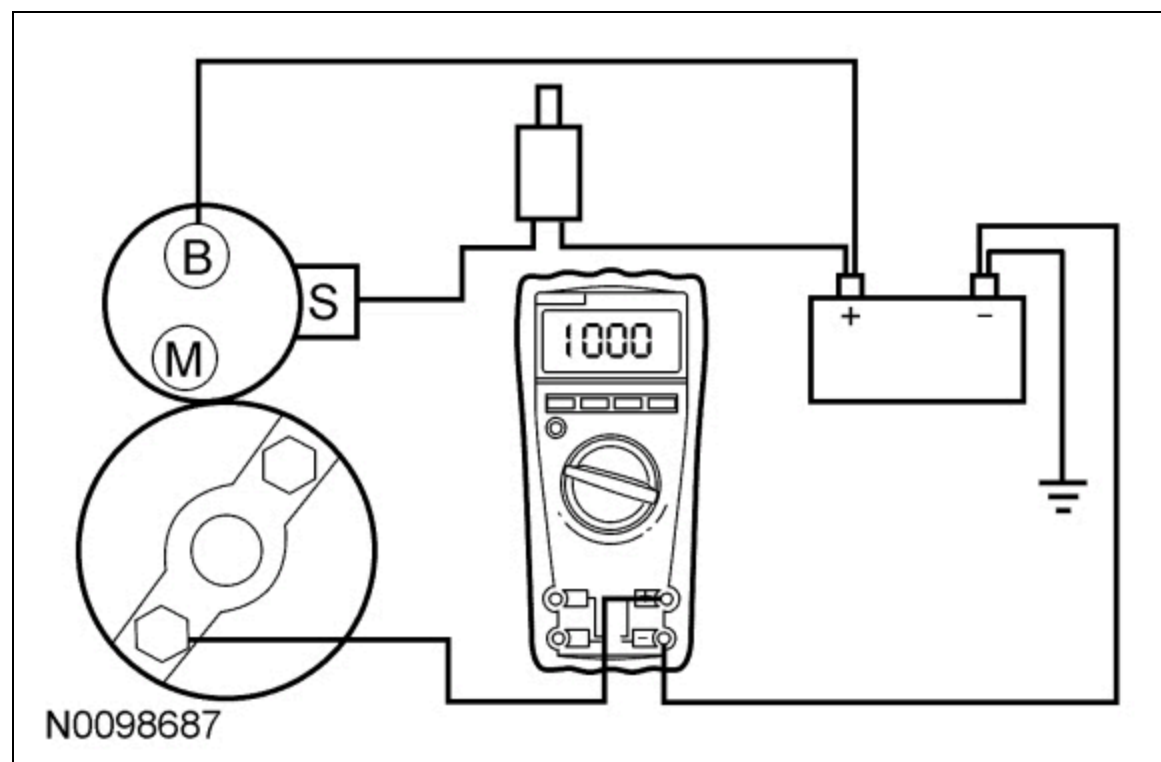
- 4. Connect the multimeter positive lead to the battery positive post. Connect the negative lead to the starter solenoid "M" terminal.
- 5. Place the gear select lever in Park or Neutral.
- 6. Engage the remote starter switch. Read and record the voltage. The voltage reading should be 0.5 volt or less.
- 7. If the voltage reading is 0.5 volt or less, go to the Starter Motor - Ground Circuit Test in this section.

8. A voltage reading greater than 0.5 volt is an indication of excessive resistance in the starter connections, the positive battery cable or in the starter solenoid. Remove the cables from solenoid "B", "S", and "M" terminals. Clean the cable connections and reinstall the cables to the correct terminals. Repeat Steps 3 through 7.
9. If the voltage reading is still greater than 0.5 volt when checked at the "M" terminal, move the multimeter negative lead to the starter solenoid "B" terminal.
10. With the gear select lever in Park or Neutral, engage the remote starter switch. Read and record the voltage.
11. If the voltage reading at the "B" terminal is lower than 0.5 volt, the concern is in the connections at the starter solenoid or in the solenoid contacts. Install a new starter motor. Refer to Starter Motor — 3.5L GTDI, 3.7L or Refer to Starter Motor — 5.0L (4V), 6.2L (2V)
12. If the voltage reading taken at the solenoid "B" terminal is still greater than 0.5 volt after cleaning the cables and connections at the solenoid, the concern is either in the positive battery cable connections or in the positive battery cable itself.
13. Clean the positive battery cable connections and retest. If this does not solve the problem, install a new positive battery cable. Section 414-01.

Starter Motor - Ground Circuit Test

A slow cranking condition can be caused by resistance in the ground or return portion of the cranking circuit. This procedure checks the voltage drop in the ground circuit.

1. Connect a remote starter switch between starter solenoid "S" terminal and the battery positive terminal.
2. Connect the multimeter positive lead to the starter motor housing (the connection must be clean and free of rust or grease). Connect the negative lead to the negative battery terminal.



3. Place the gear select lever in Park or Neutral.
4. Engage the remote starter switch and crank the engine. Read and record the voltage reading. The reading should be 0.5 volt or less.
5. If the voltage drop is greater than 0.5 volt, clean the negative cable connections at the battery, the body ground connections and the starter ground connection. Retest.

6. If the voltage drop is greater than 0.5 volt, install a new negative battery cable. Section 414-01.
7. If the voltage reading is less than 0.5 volt and the engine still cranks slowly, install a new starter motor. Refer to Refer to Starter Motor — 3.5L GTDI, 3.7L or Refer to Starter Motor — 5.0L (4V), 6.2L (2V).